# Airbnb Cape Town Business Analysis.

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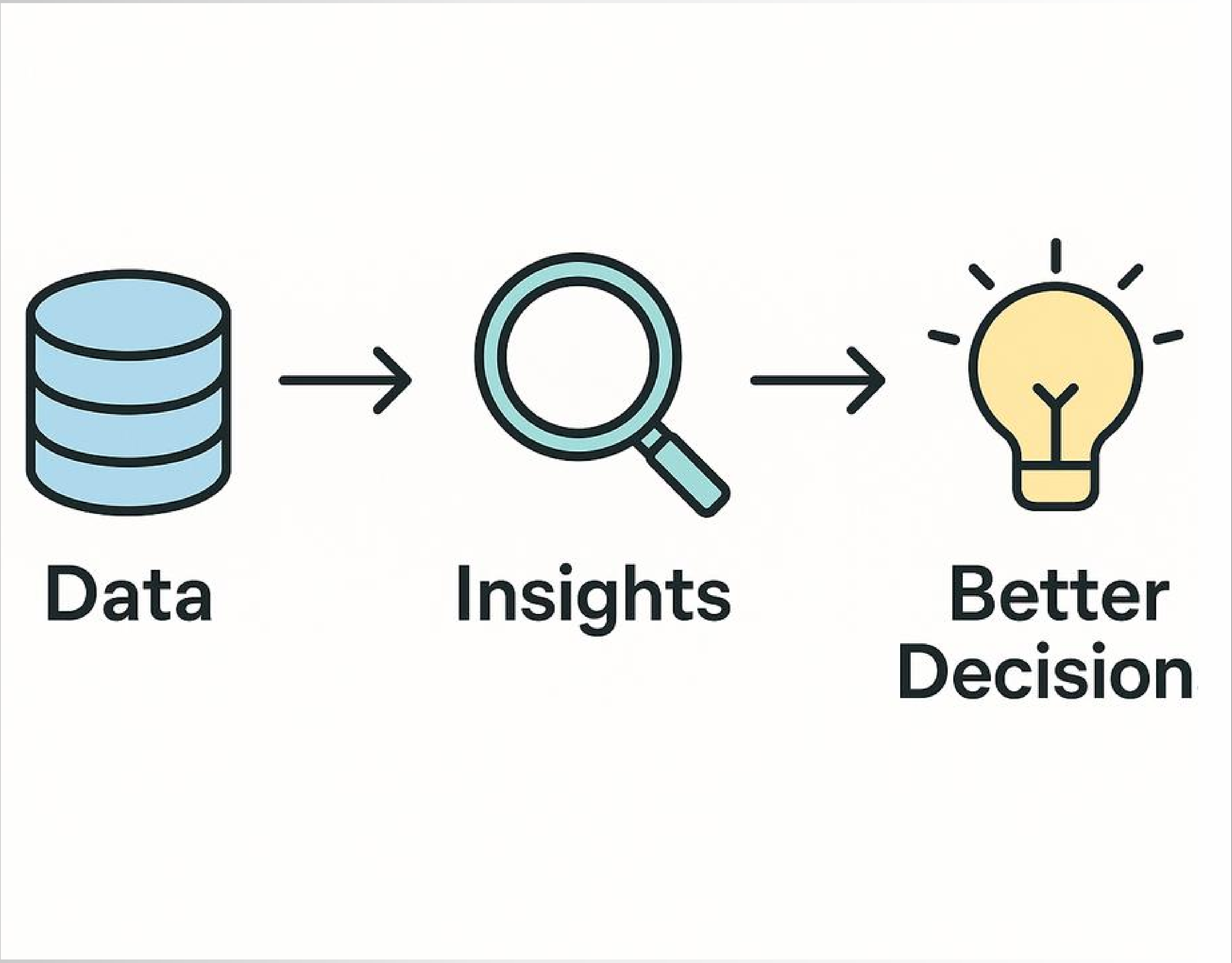
A data-driven guide for Airbnb hosts

# OVERVIEW

- Cape Town’s Airbnb market is competitive and fast-growing.
- Many hosts—especially new ones—struggle to set prices and estimate revenue.
- We built a data-driven tool to help hosts:

  -Predict expected revenue.

  - Suggest improvements to increase earnings.



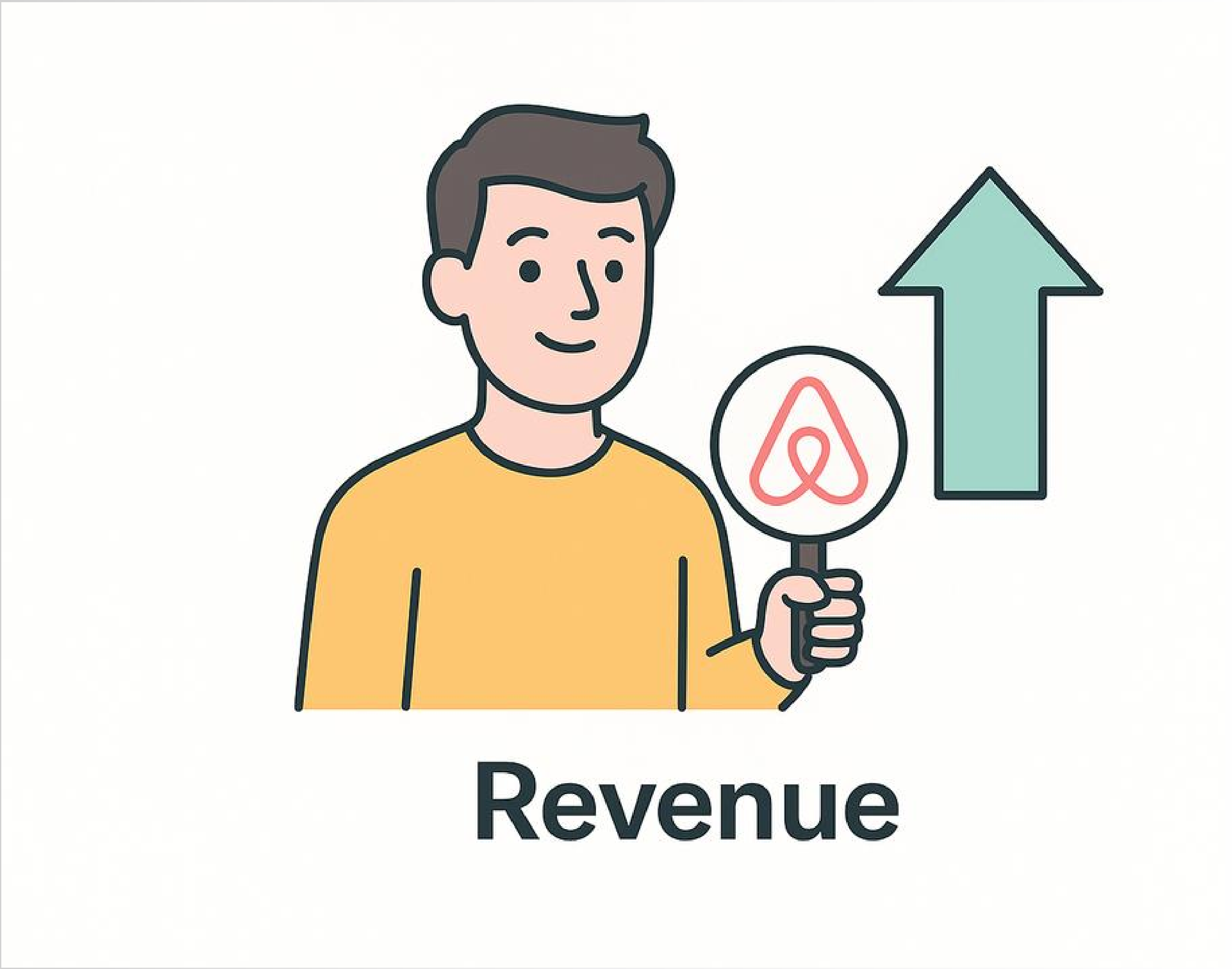
# BUSINESS UNDERSTANDING

The Problem:
- Hosts risk losing bookings or undervaluing their property without good pricing and availability strategies.

Our Solution:
- A Revenue Recommender System that learns from past Airbnb data and gives clear, actionable tips.

Goal:
- Identify drivers of revenue

Predict expected revenue based on price, availability, location, and reviews."



# Data Understanding

## Data Sources

We used Airbnb's public Cape Town datasets:

- **Listings.csv** – Property details (location, room type, amenities)
- **Calendar.csv** – Daily prices and availability
- **Reviews.csv** – Guest ratings and comments
- **Neighbourhoods.csv** – Mapping of areas in Cape Town

# Data Preparation

To make sure our analysis was accurate, we:

- Removed outliers
- Filled missing values in reviews and availability.
- Converted categories to numbers so models can understand them.
- Created new features like seasonal availability trends.

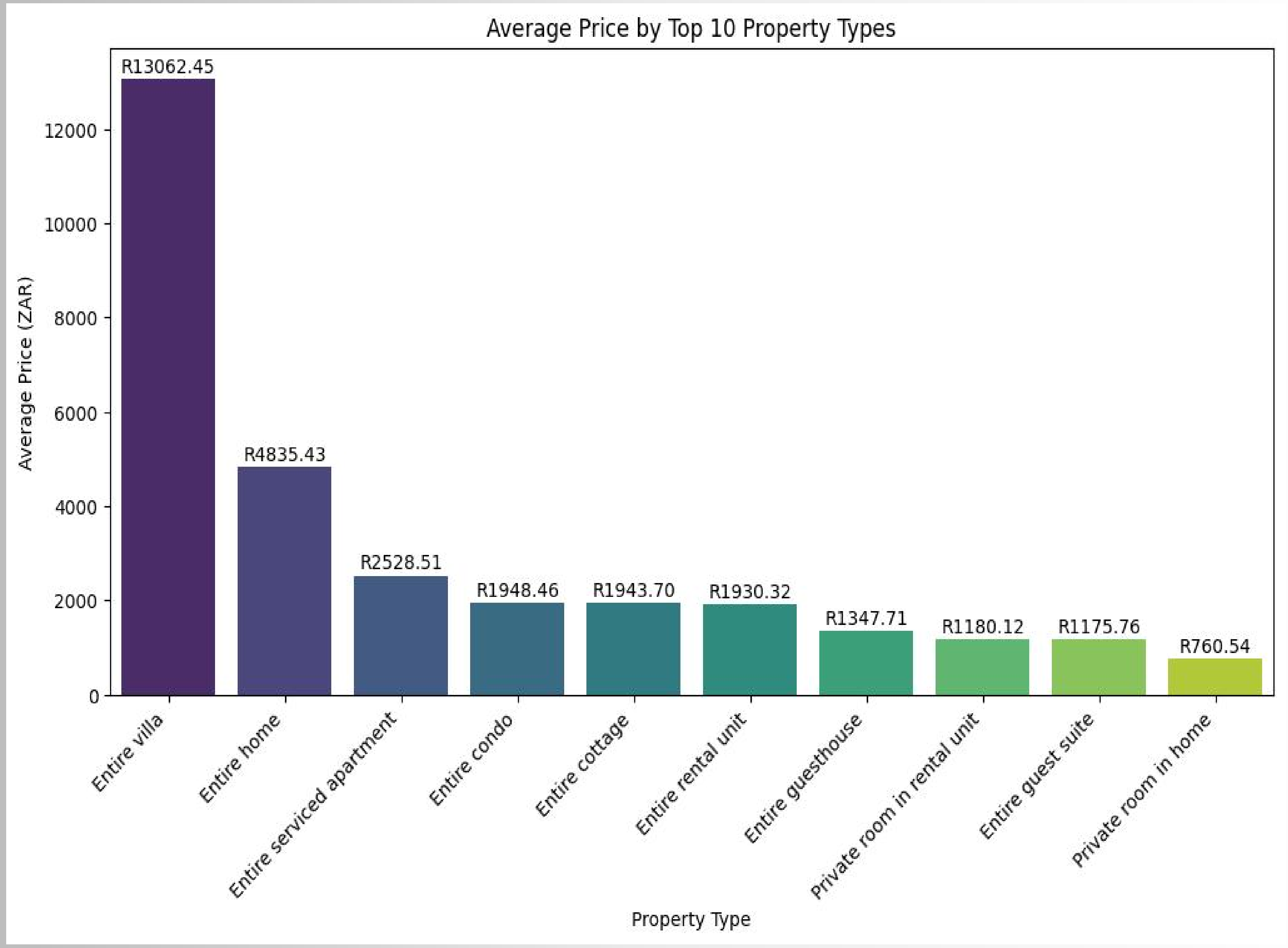
# Key Data Insights

property type has a huge impact on earnings.

- Luxury properties earn far more: Villas charge the highest average price
- Entire homes are much more profitable (~R4,800) compared to smaller units.

- Serviced apartments and condos are mid-range options (~R2,000–R2,500).

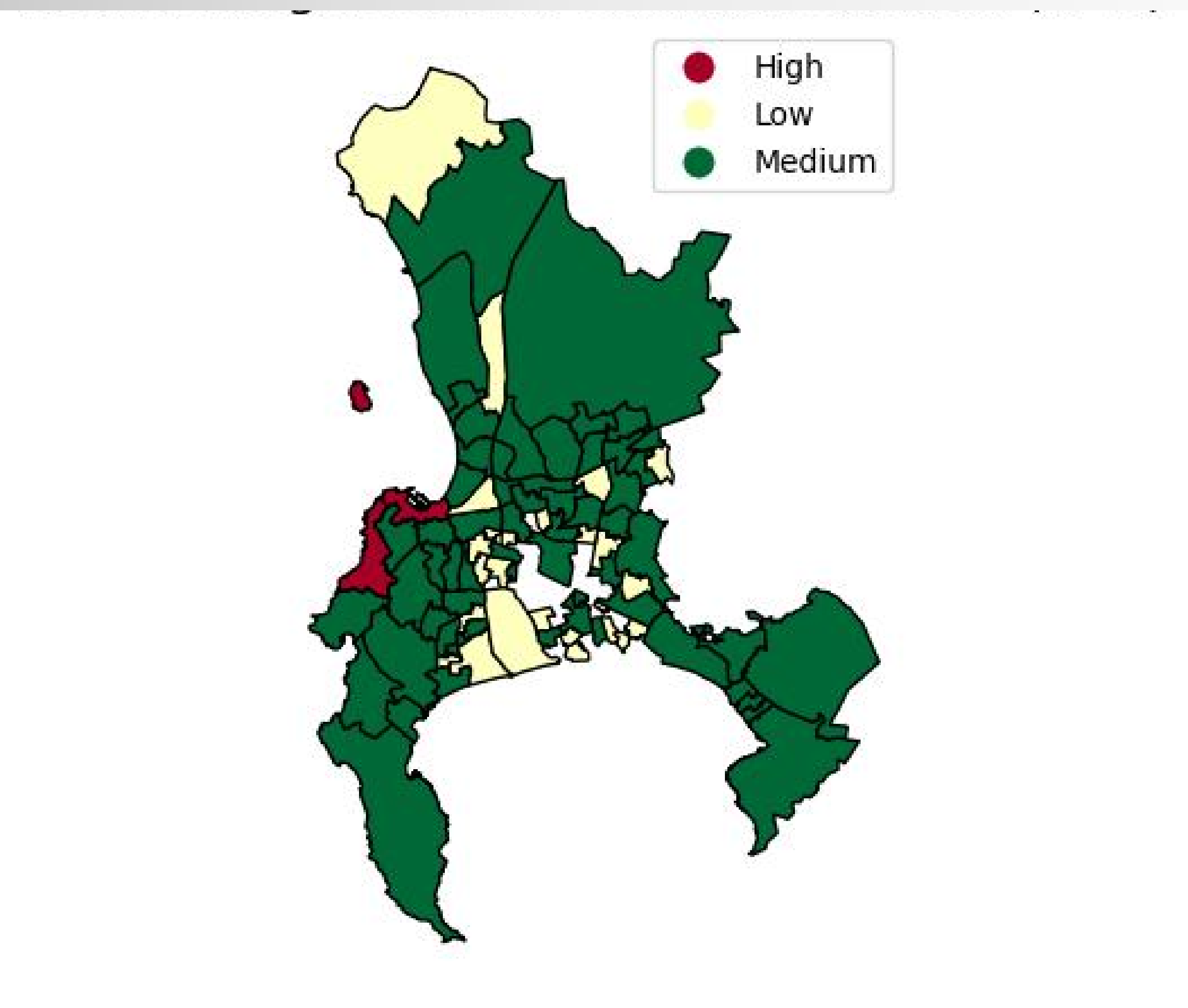
- Private rooms earn the least (under R1,200 on average).



# GEOGRAPHIC INSIGHTS

Airbnb Neighborhood Demand

- High-demand zones (Red): concentrated in central/west areas with strong tourism pull.
- Medium-demand zones (Green): cover most neighborhoods, showing steady but moderate bookings.
- Low-demand zones (Yellow): scattered pockets, often peripheral or less connected.



# Modelling

We tried several "smart tools" to see which predicts revenue best:

- Linear Regression – Looks for straight-line relationships between factors like price, location, and revenue.
- Random Forest – Handles complex situations where many factors work together.
- Gradient Boosting – Combines many small models to make very accurate predictions.
- Clustering – Groups similar listings together so we can compare performance within each group.

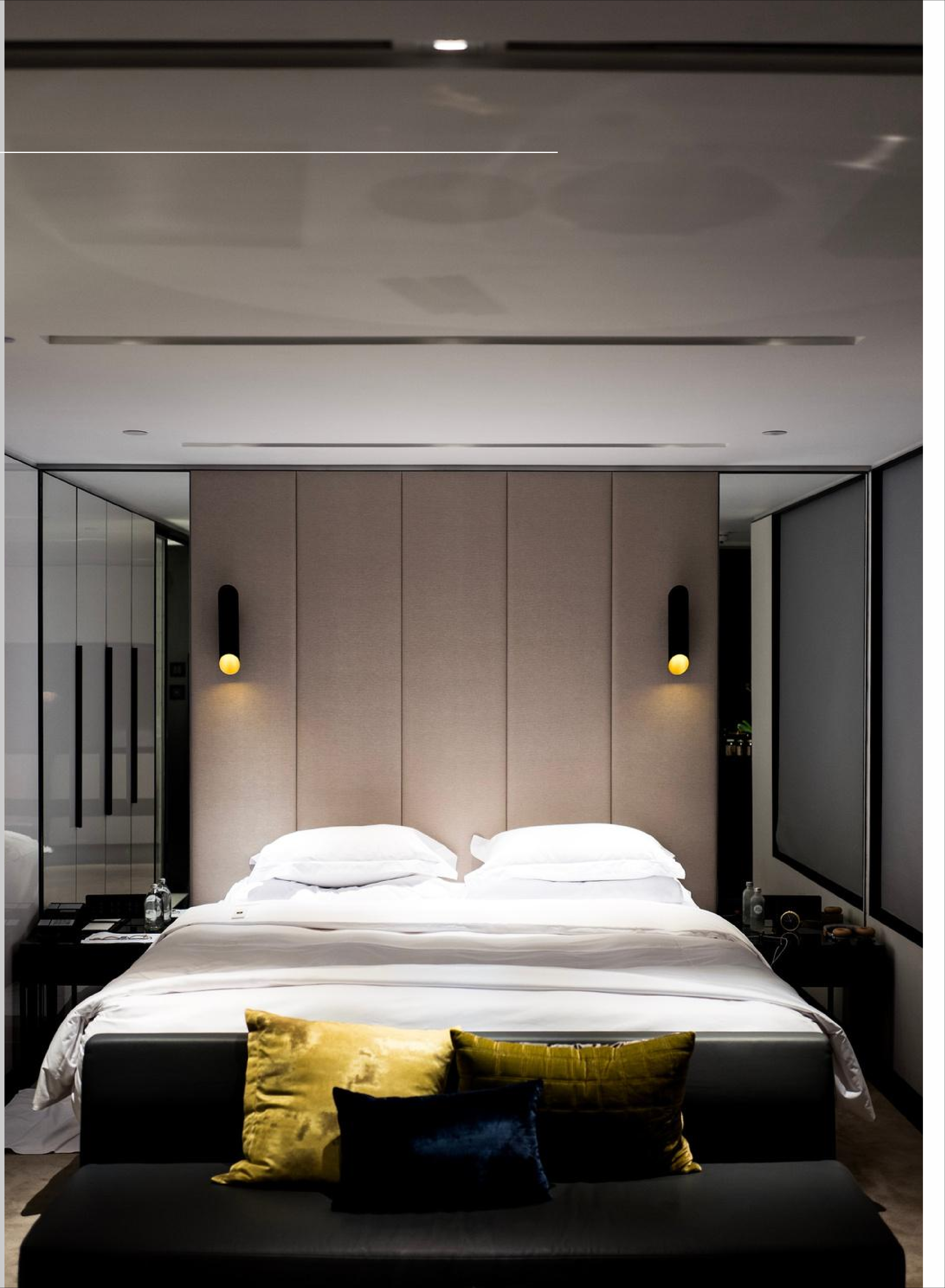
# Findings

- Price, location, and room type are the biggest drivers of revenue
- Higher review scores boost revenue potential
- Availability patterns (especially during peak season) matter a lot

# Recommendations

To boost revenue, hosts should:

- Offer entire homes if possible
- Increase booking availability
- Maintain high review scores
- Optimize pricing using our model’s feedback



# Next Steps

We plan to:

✅ Launch a prototype dashboard using Streamlit

✅ Allow host input + instant feedback

✅ Deploy model as an API for future apps

✅ Share with local host communities

# Limitations

- Data may not reflect recent Airbnb trends
- Sparse reviews or availability can reduce accuracy
- External market shocks are not captured in this dataset.

# Conclusion

- To wrap up ,our tool empowers hosts with clear, data-backed insights to improve earnings and stay competitive. We’d be happy to dive deeper into any part of the analysis.

# THANK YOU

## Questions and feedback

**We are ready to discuss Further**